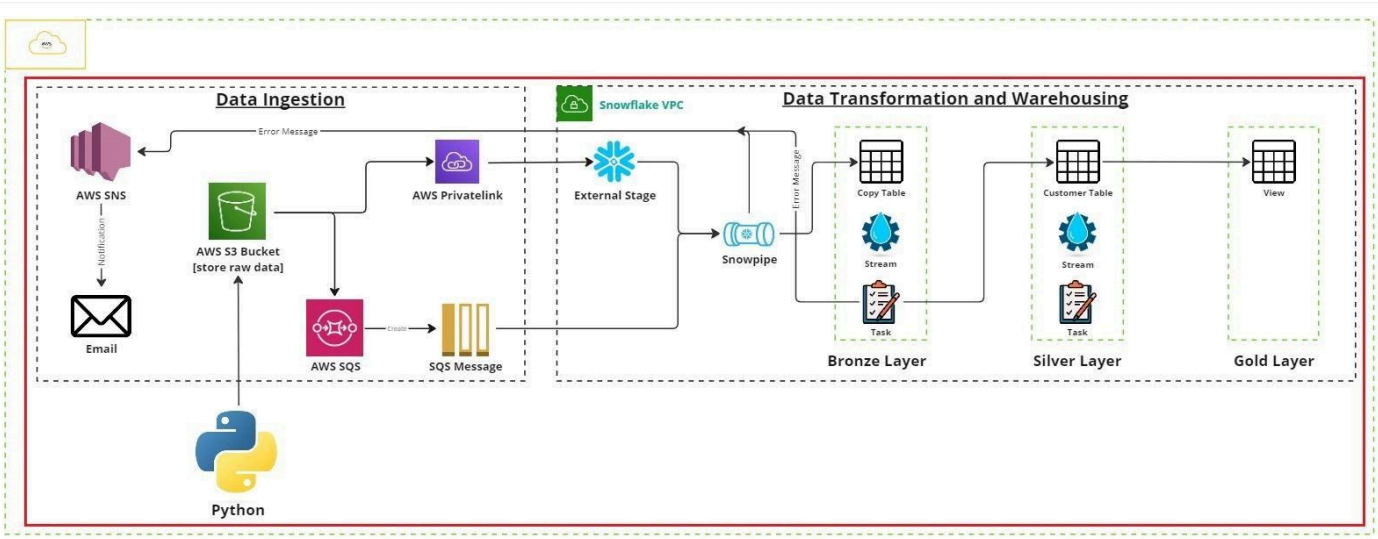
Technical Overview

In this document, we are going to test the end to end setup of the SCD1 pipeline. Also verify that the SCD1 logic is working as expected based on the Primary key of the customer dataset.

Architecture



Pre-requisites:

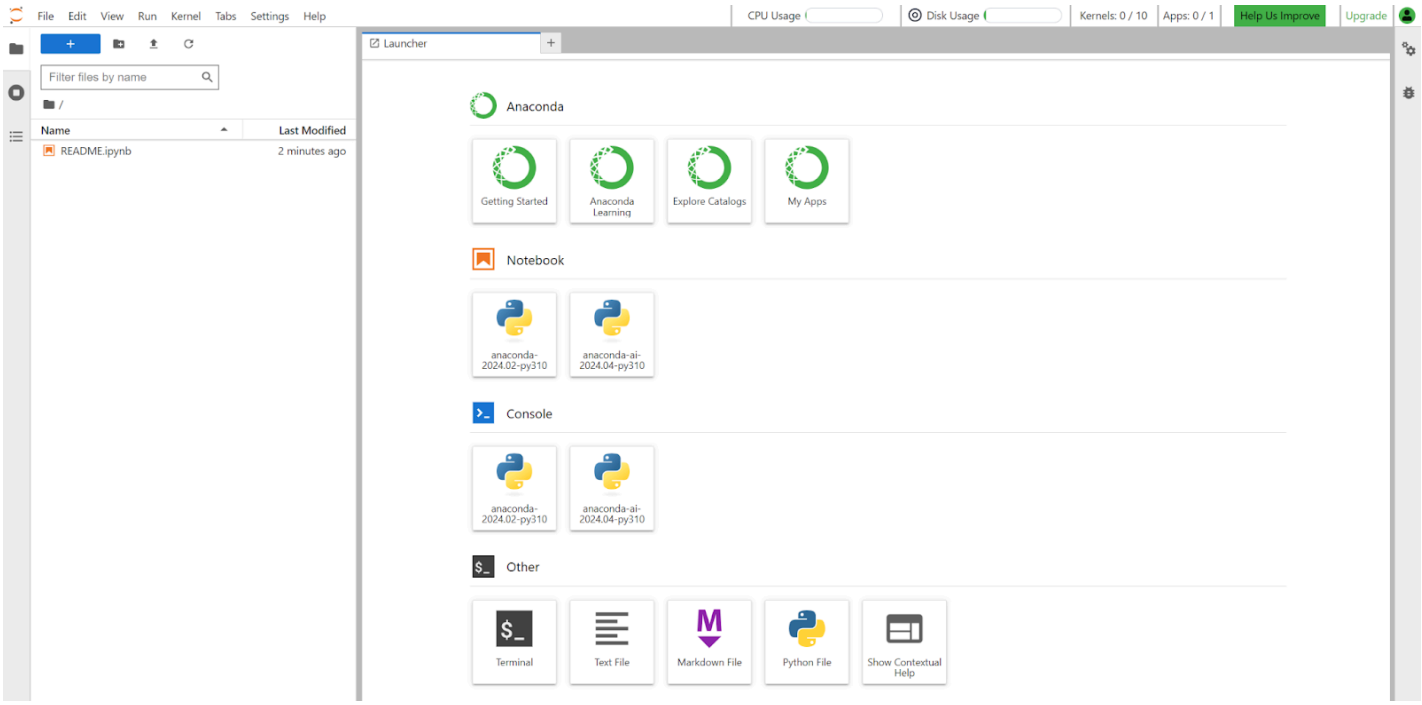
- Notebook Setup in Anaconda Cloud
- AWS Account
- Snowflake Account

Download the Customer Dataset

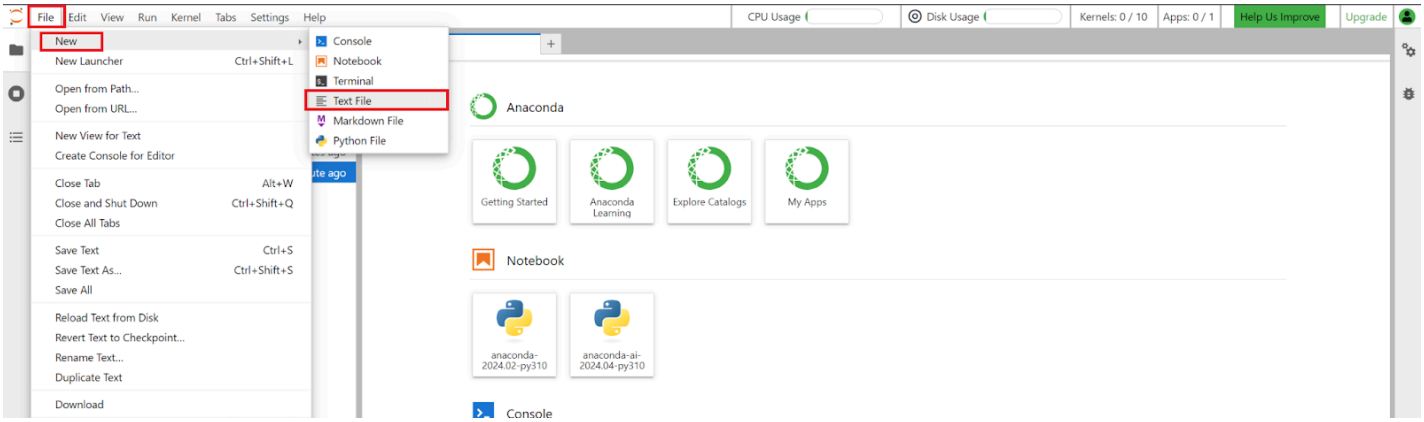
- Download the provided customer dataset csv files as part of this project to your local machine.
- There will be two datafiles shred as part of this project
  - customer_full_data.csv : This is the csv file containing the entire customer dataset
  - customer_change_data : This is the csv file containing few customers updated records along with few new customer records which are not present in the customer_full_data.csv file

Setup the yml file in anaconda cloud to store the AWS Access/Secret keys and region

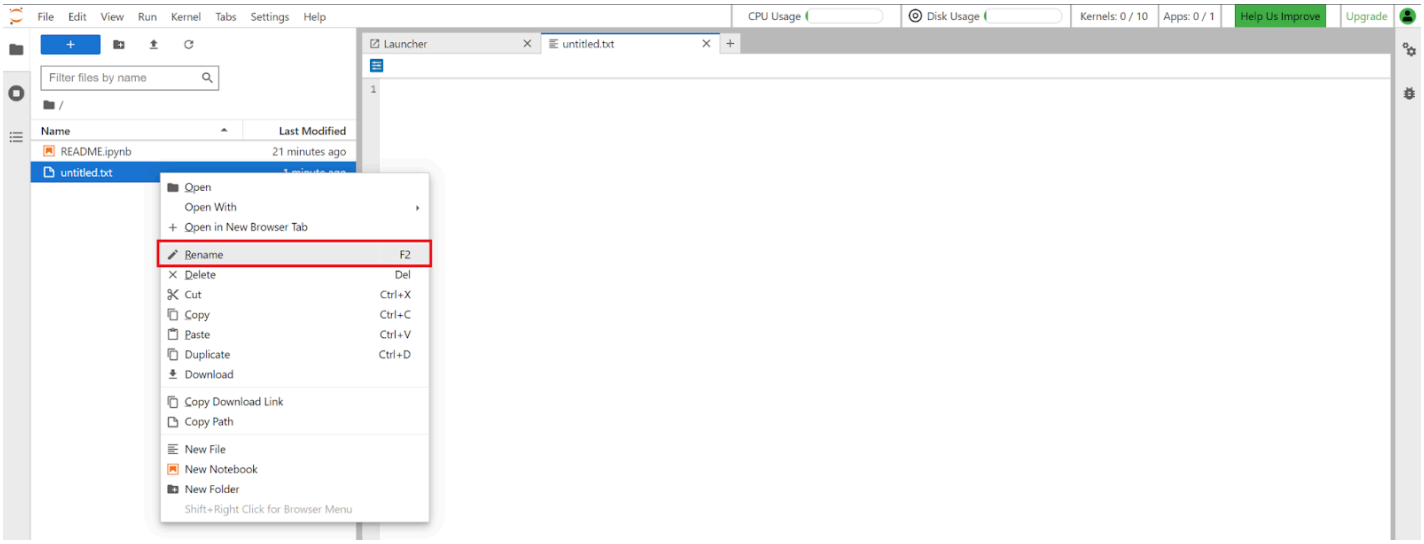
- Open you anaconda cloud notebook page which you have already bookmarked as part of the previous steps



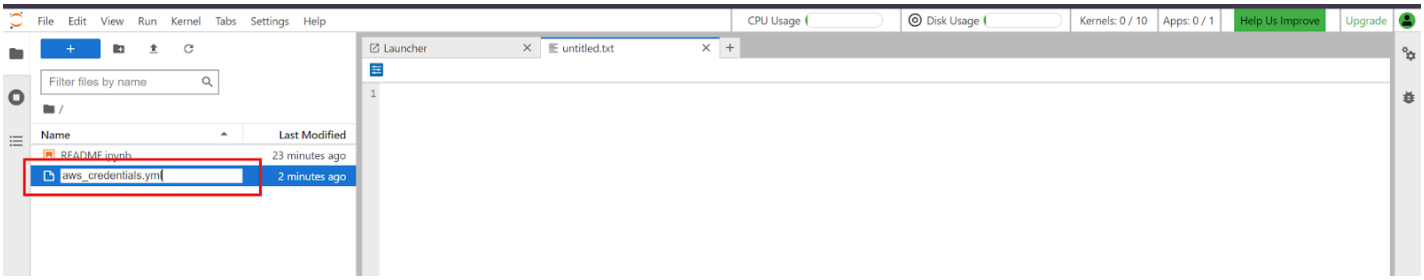
- Click on File option, under File select New and then select Text File



- Right click on the newly created file and select the Rename option.



- Name the file as aws_credentials.yml

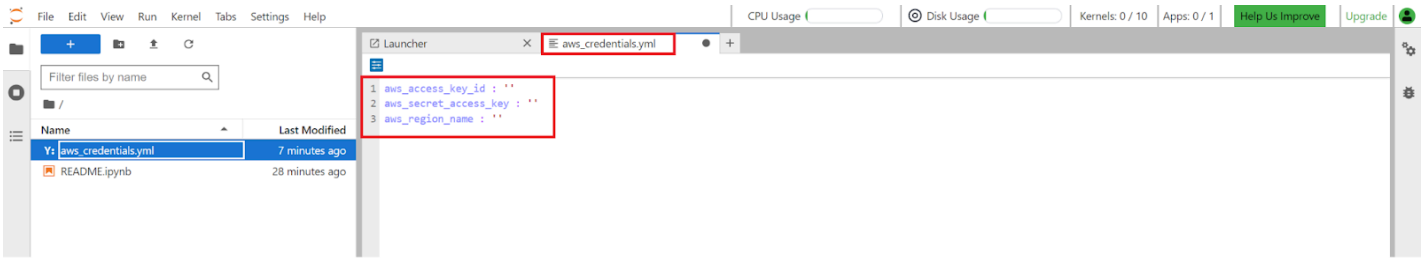


- Paste the below details in the above created aws_credentials.yml file

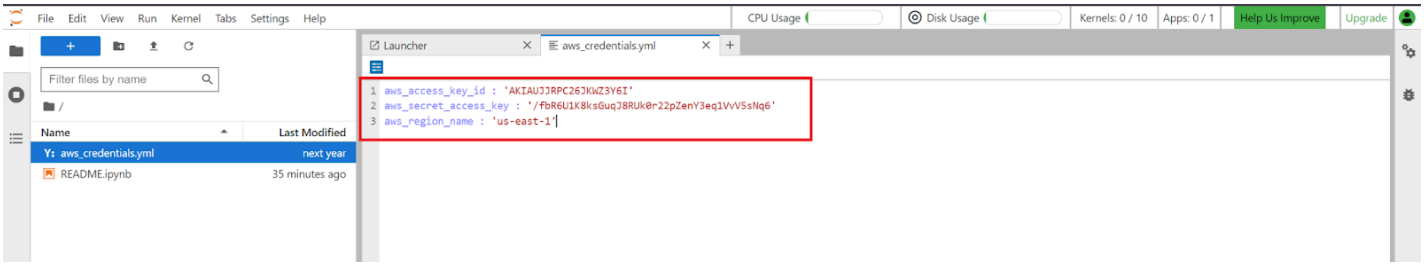
aws_access_key_id : ''

aws_secret_access_key : ''

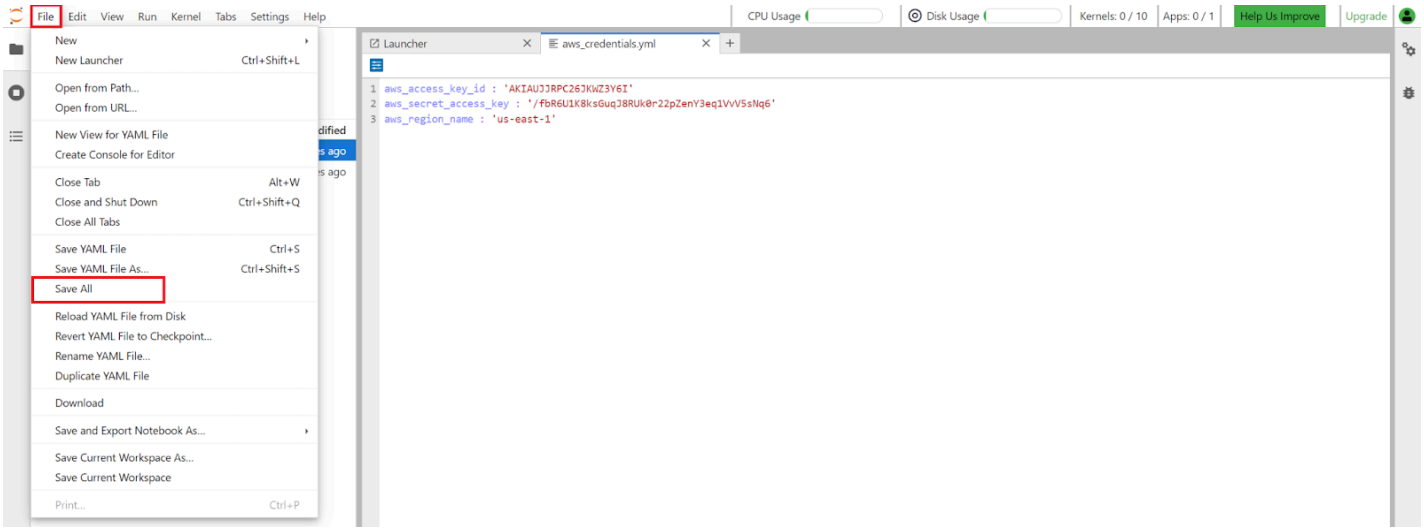
aws_region_name : ''



- Update the aws_access_key_id and aws_secret_access_key values from the key files which you have downloaded as part of the AWS IAM User creation step. The aws_region_name will be the default region where your AWS account is setup-ed.

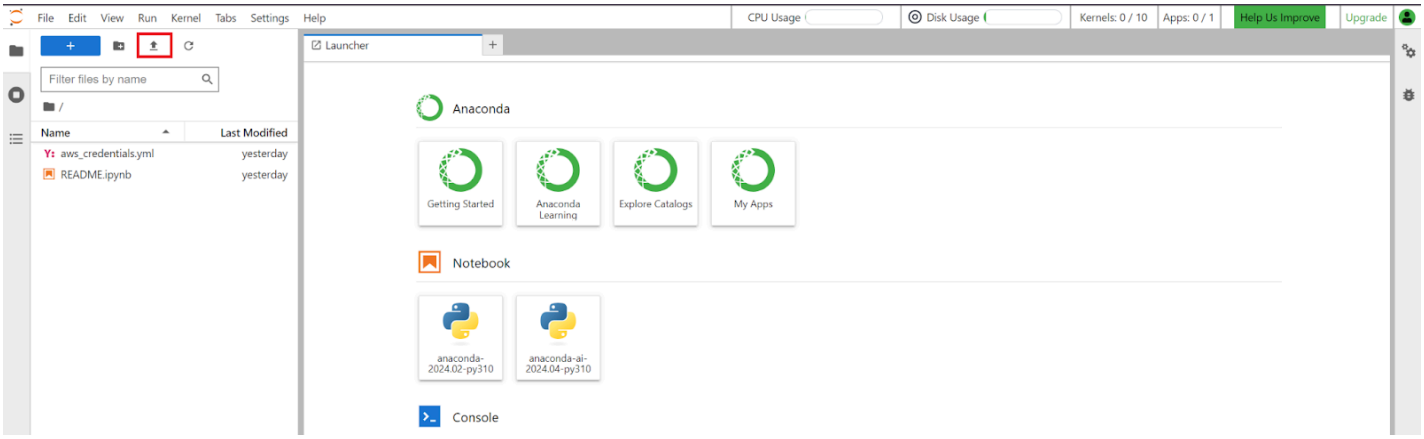


- Once the file is updated, save the file by clicking the File option and then click the Save All option.

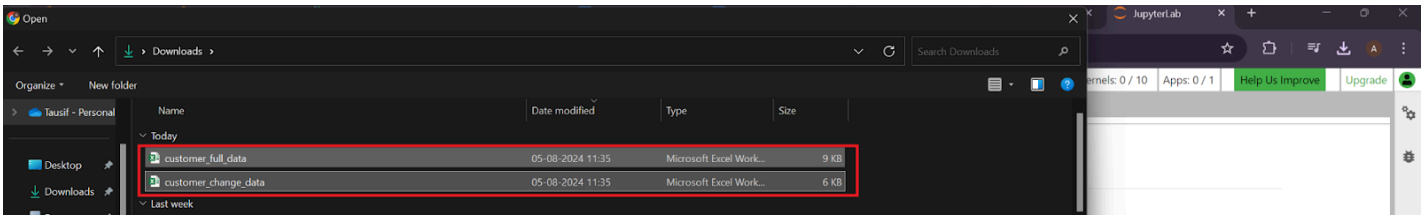


Upload the customer data file to anaconda cloud

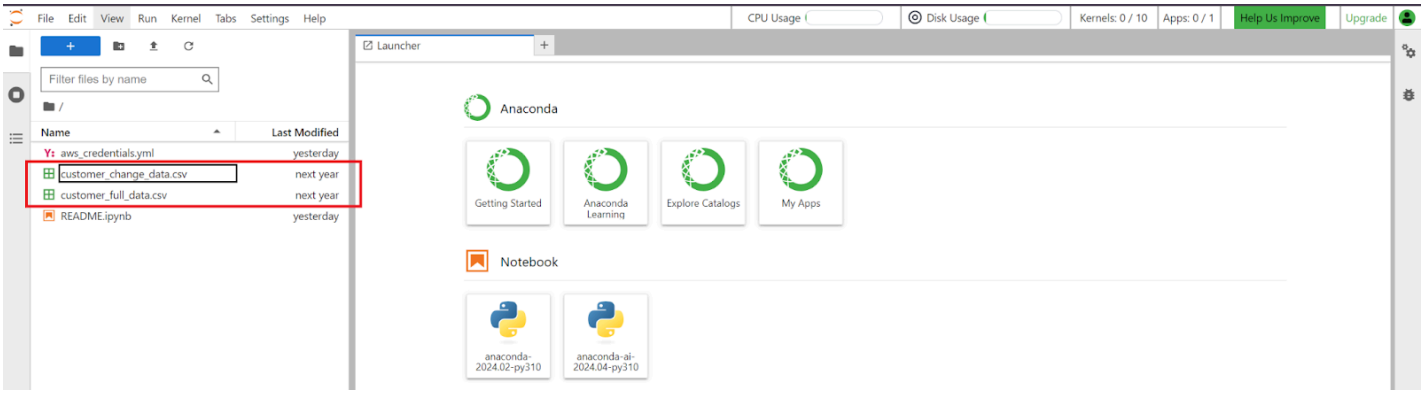
- Click on the upload button present on the left side of anaconda notebook



- Select both of the customer dataset files which you have downloaded in the previous step and upload them to your anaconda notebook

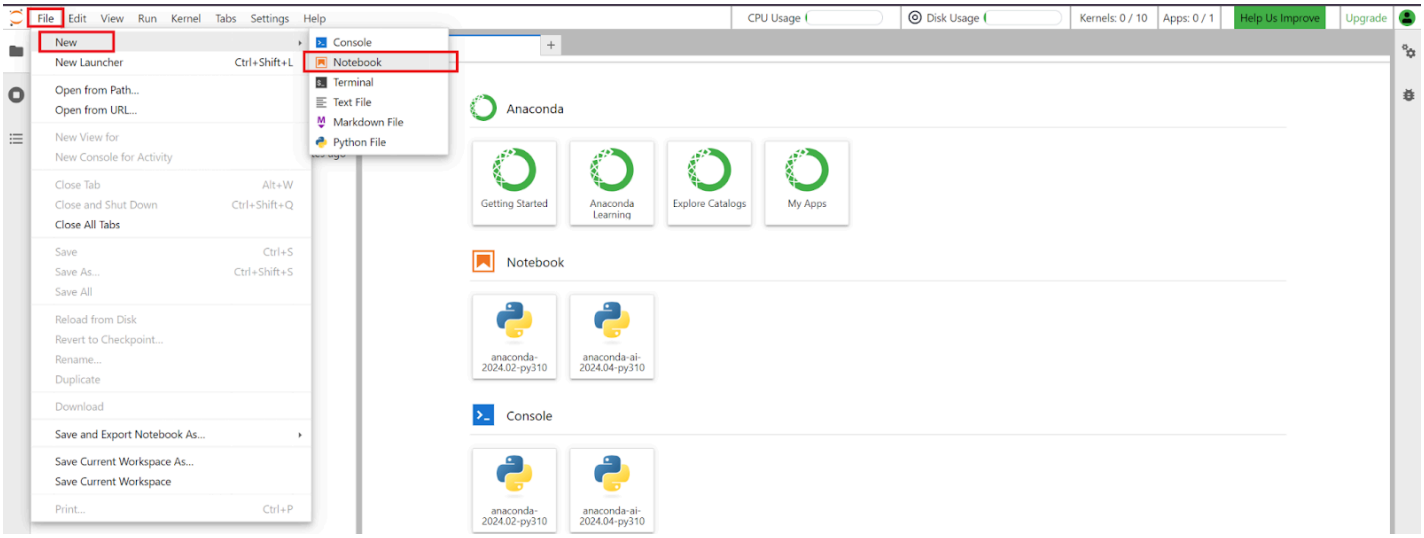


- Both of the customer data files will be successfully uploaded to your anaconda notebook for further processing.

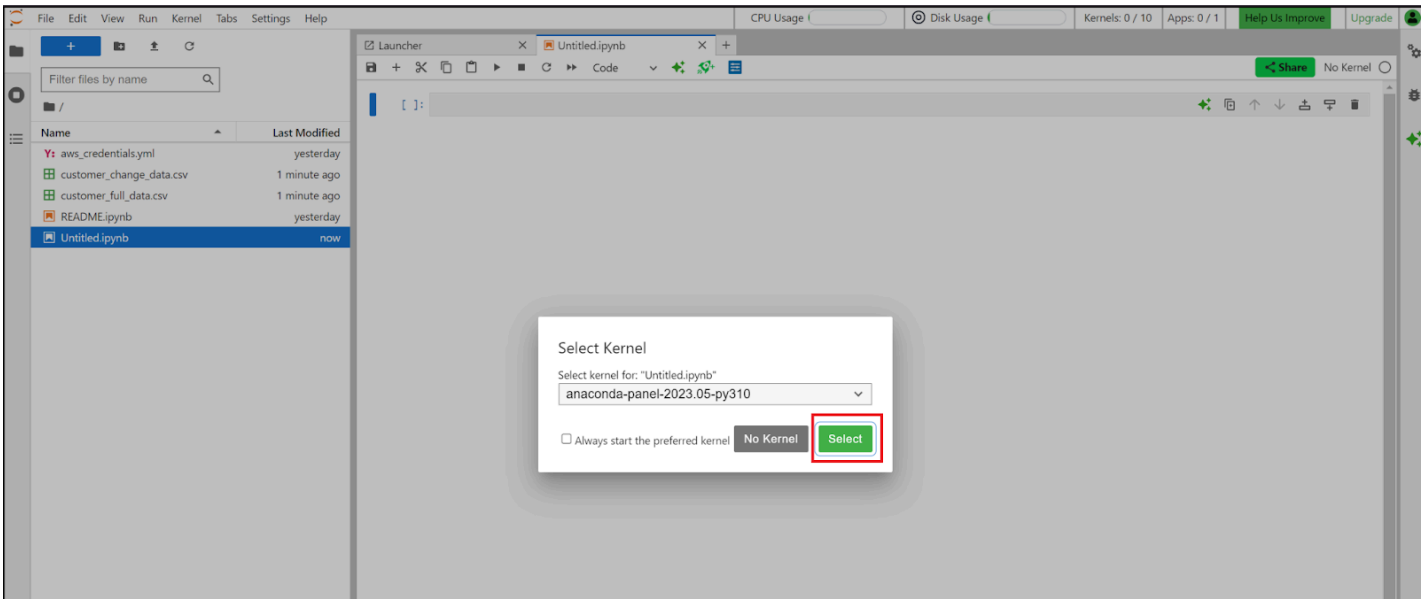


Setup the notebook in anaconda cloud to upload customer data file to S3 bucket

- Click on File option again, under File select New and then select Notebook

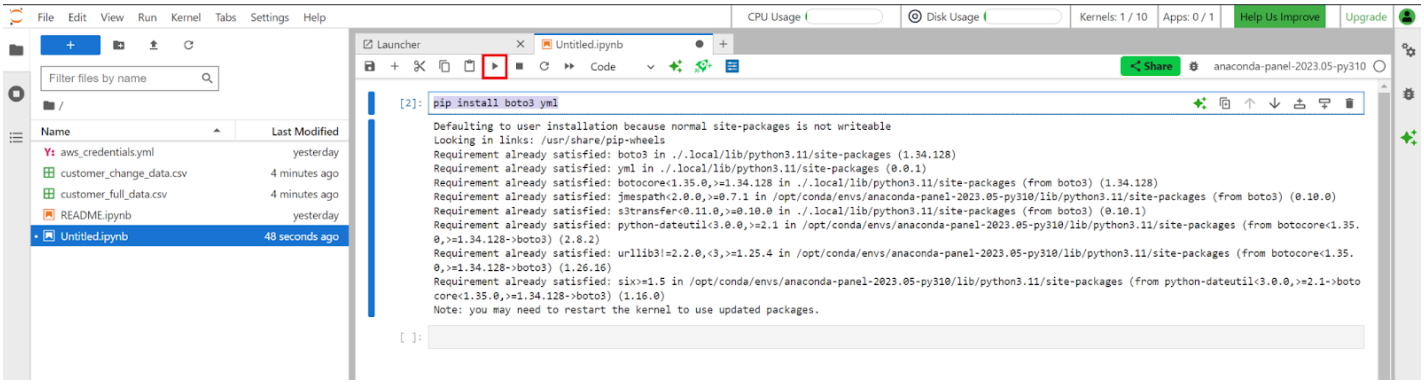


- Keep the default kernel and click on the Select button to create the notebook

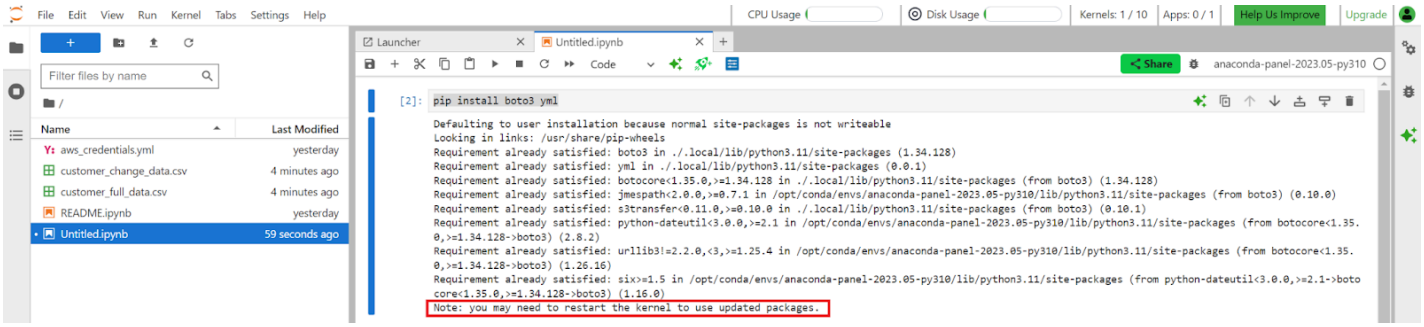


- Execute the below command in the cell of the newly created notebook by clicking the run button present on the top menu of your notebook to install the boto3 and yml libraries required for the python script

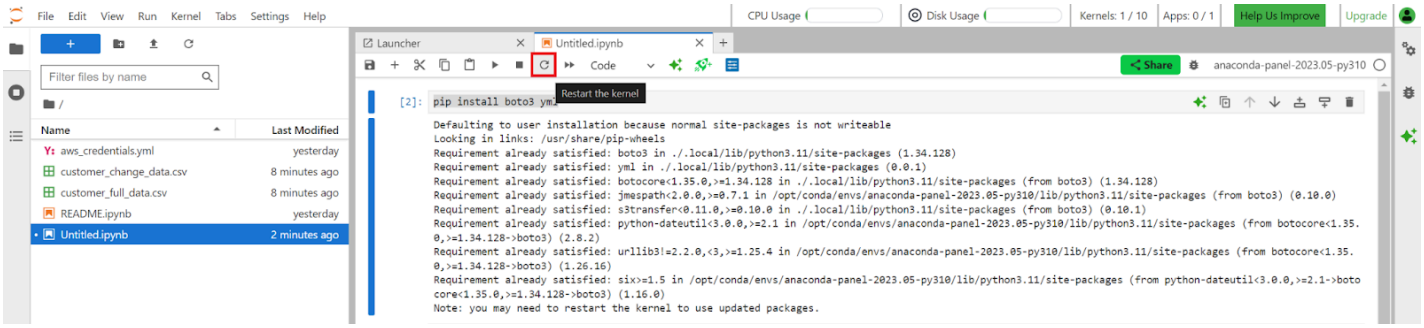
pip install boto3 yml

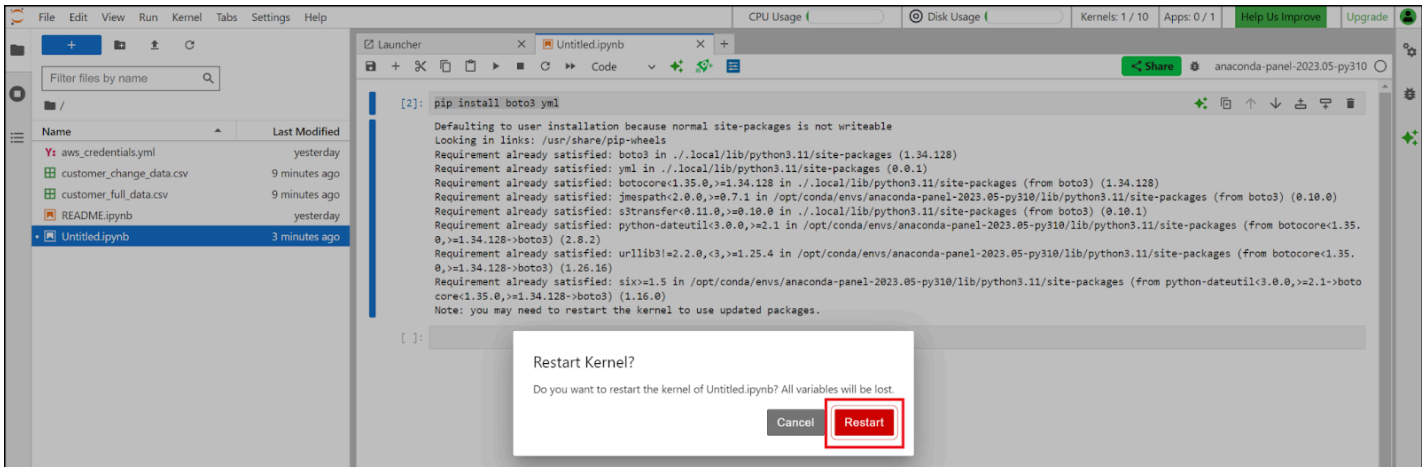


- Once successful, your boto3 and yml libraries are installed successfully you will see a message “you may need to restart the kernel to use updated packages”.



- Restart your kernel by clicking the restart button present on the top menu of your notebook





- Once restarted, paste the below python code in the cell of the notebook.

```
import boto3
import yaml

with open('aws_credentials.yml', 'r') as file:
    aws_credentials = yaml.safe_load(file)

# Extract credentials from YAML
aws_access_key_id = aws_credentials['aws_access_key_id']
aws_secret_access_key = aws_credentials['aws_secret_access_key']
aws_region_name = aws_credentials['aws_region_name']

# Initialize a session using AWS credentials
session = boto3.Session(
    aws_access_key_id=aws_access_key_id,
    aws_secret_access_key=aws_secret_access_key,
    region_name=aws_region_name
)

#Update the below variables values with your details
```

```
file_name = ''

bucket_name = ''

folder_name = ''


# Create an S3 client

s3_client = session.client('s3')


try:

    # Upload file in S3

    s3_client.upload_file(

    Filename =file_name,

    Bucket=bucket_name,

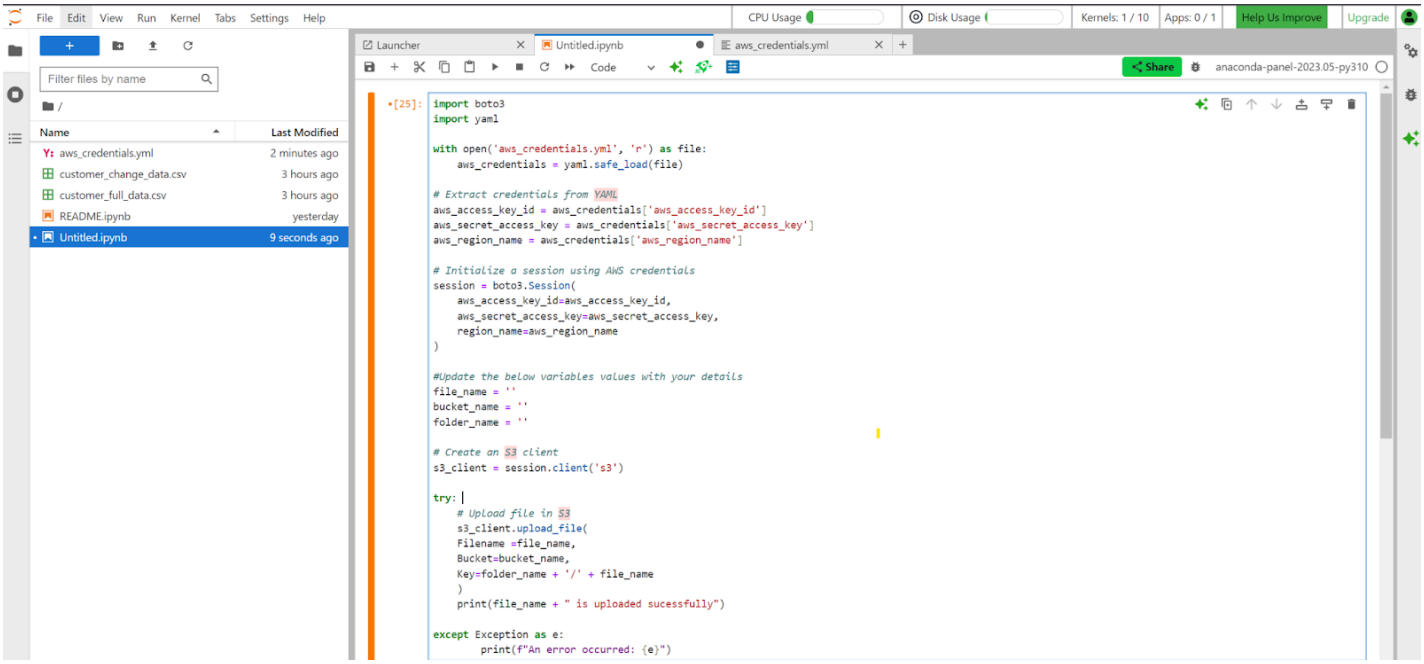
    Key=folder_name + '/' + file_name

    )

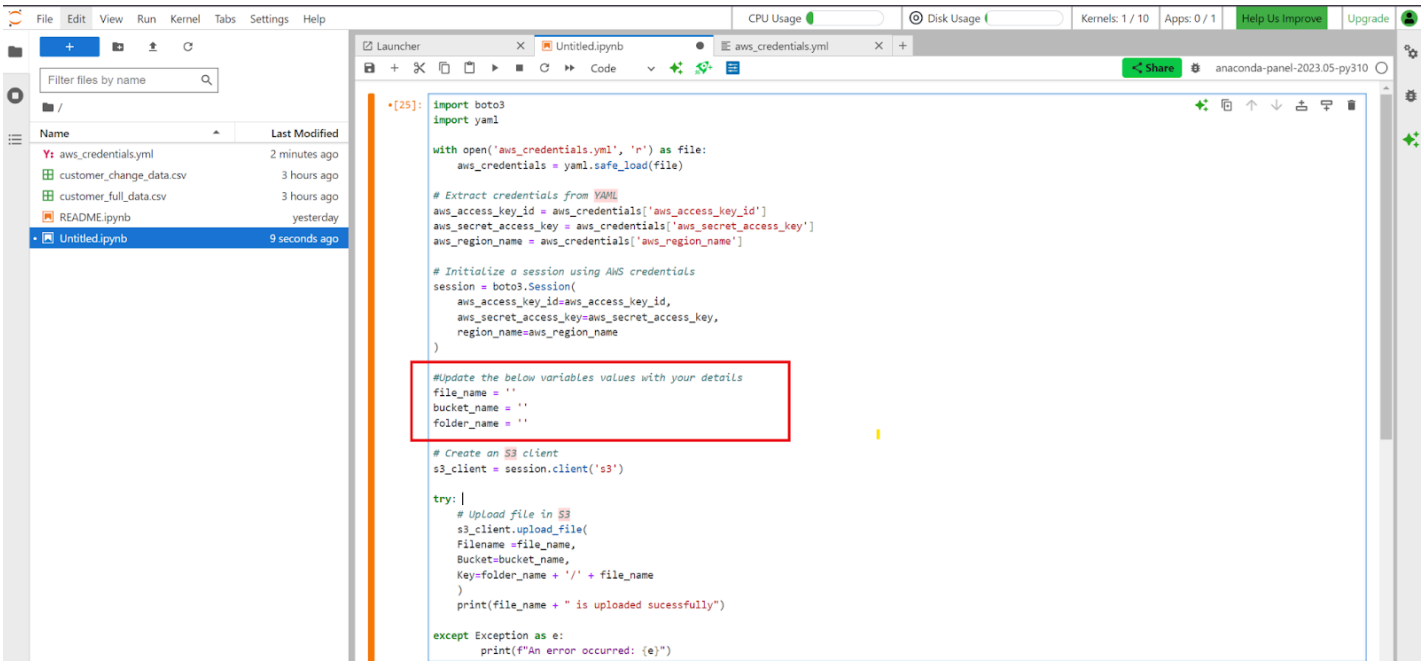
    print(file_name + " is uploaded sucessfully")


except Exception as e:

        print(f"An error occurred: {e}")
```



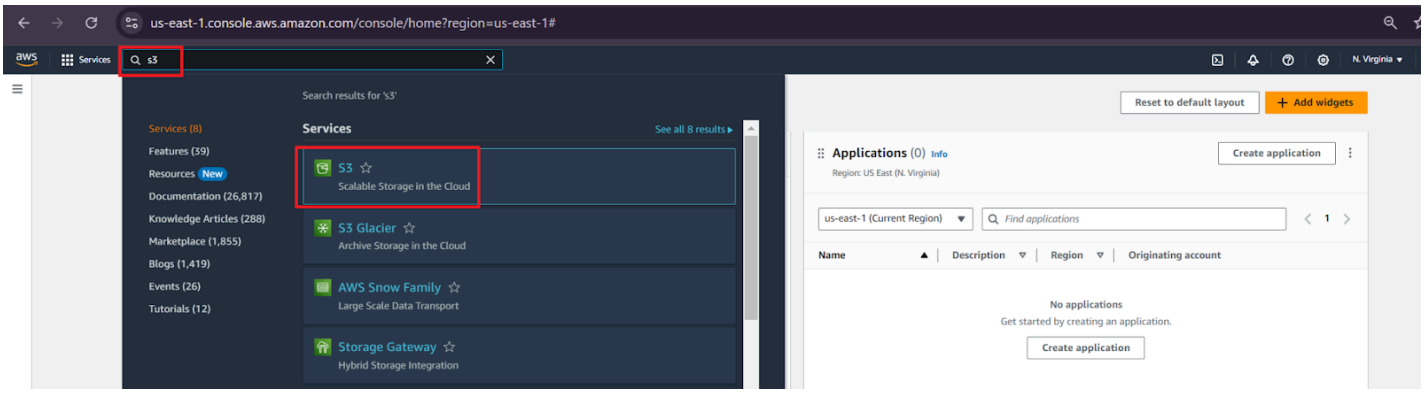
- Update the values of the below highlighted variables based on your details. For the first execution we will try to upload the customer_full_data.csv file in S3 bucket



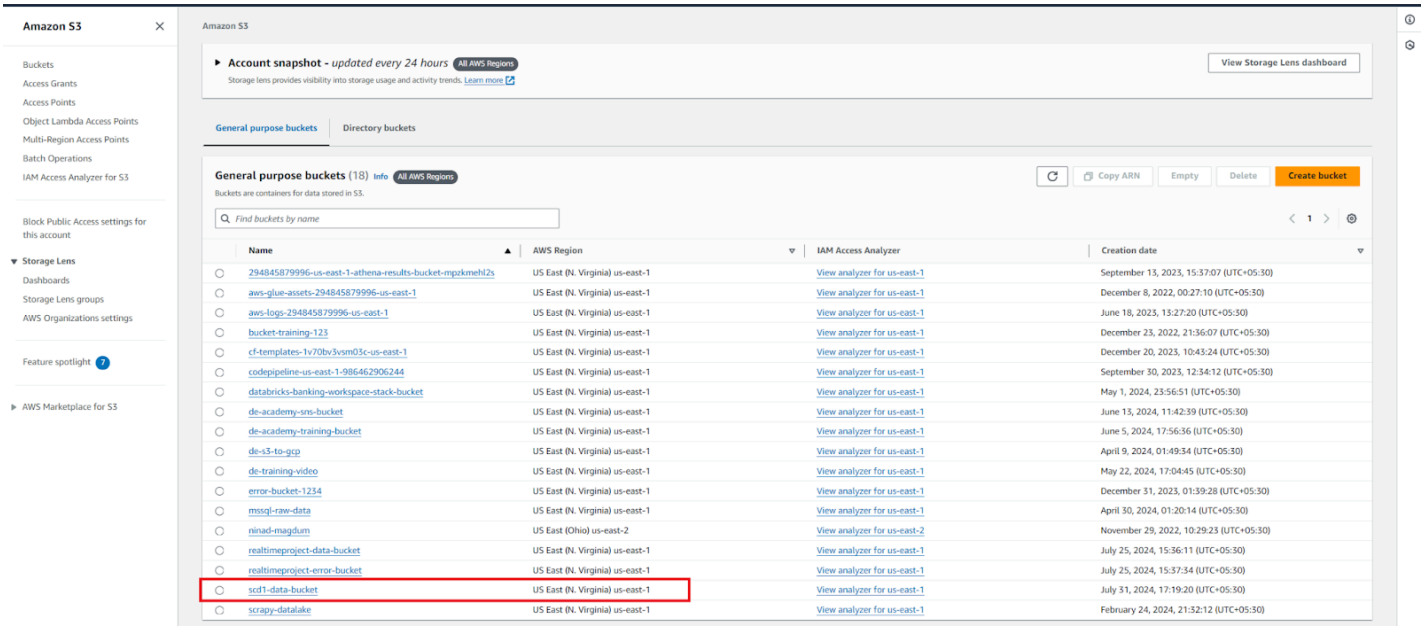
- After updating the values of the variables based on your details, click on the run button present on the top menu of your notebook to upload the customer_full_data.csv file in your respective S3 bucket folder.

- After clicking on the run button, you should be able to see the message as customer_full_data.csv is uploaded successfully

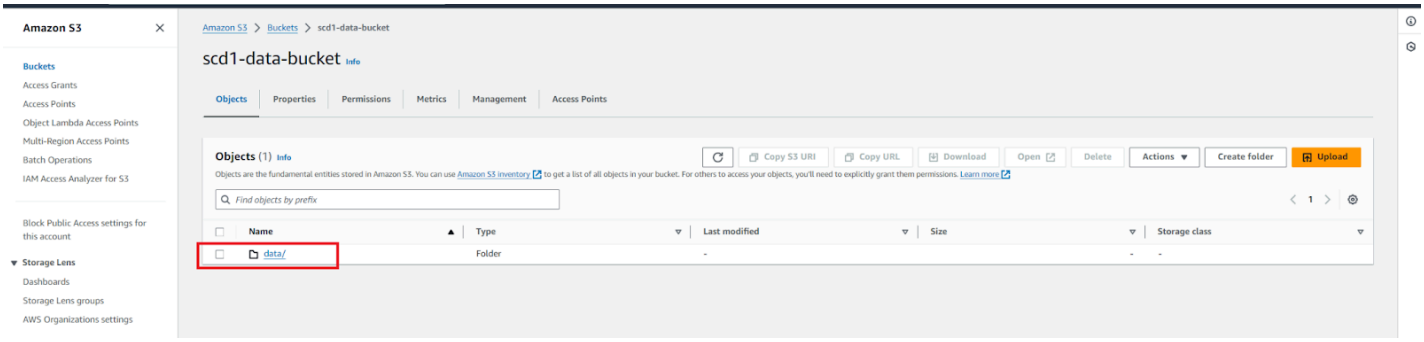
- Now the customer_full_data.csv file should be successfully uploaded to the S3 bucket folder. To validate please open the S3 bucket your AWS console by searching s3 in the search bar.



- Open you bucket once you are in the s3 bucket page



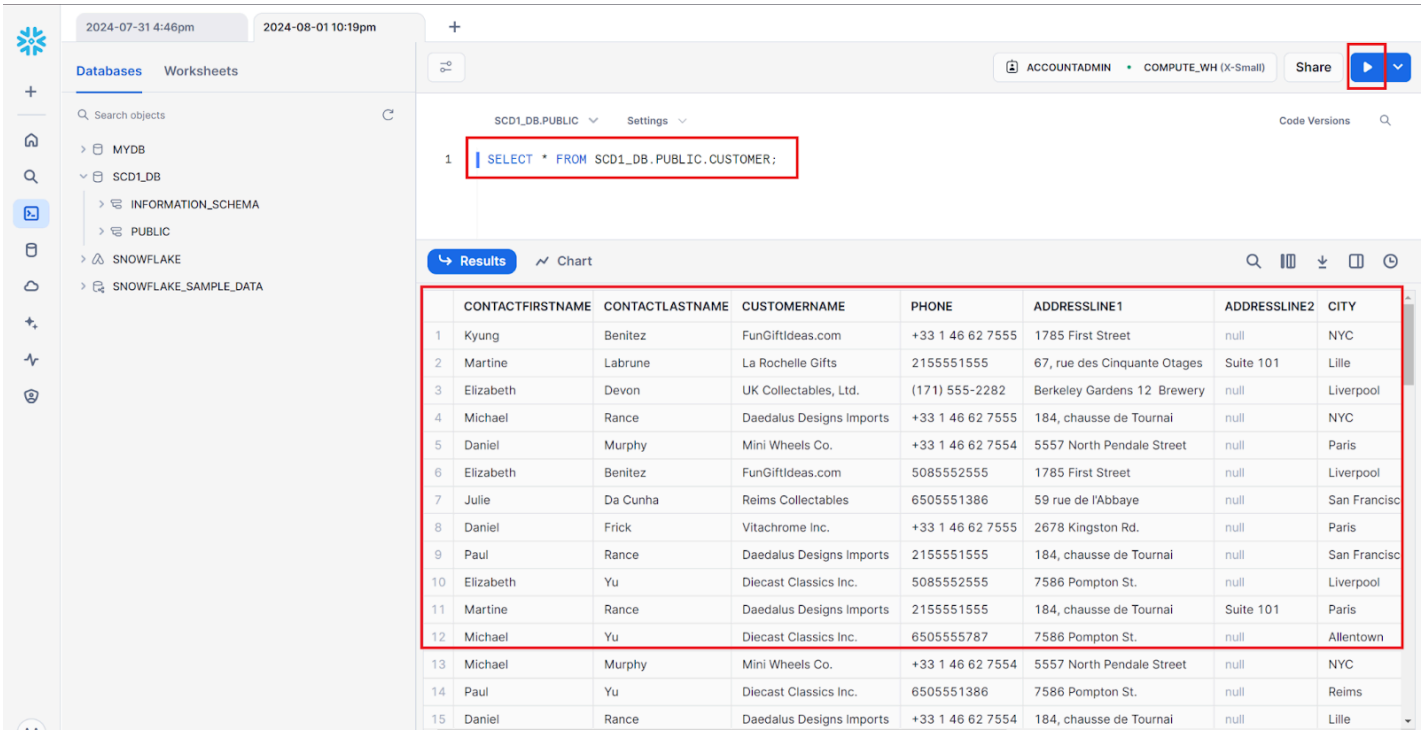
- Open the folder inside the above bucket



- Once you open the above folder, you should be able to see customer_full_data.csv file inside the folder.

- Now let’s check the customer target table after 1 min in the snowflake to confirm if the data is now successfully loaded in the table by running the below SQL query in the snowflake worksheet. You should see 80 records.

SELECT * FROM SCD1_DB.PUBLIC.CUSTOMER;



	CONTACTFIRSTNAME	CONTACTLASTNAME	CUSTOMERNAME	PHONE	ADDRESSLINE1	ADDRESSLINE2	CITY
1	Kyung	Benitez	FunGiftIdeas.com	+33 1 46 62 7555	1785 First Street	null	NYC
2	Martine	Labrune	La Rochelle Gifts	2155551555	67, rue des Cinquante Otages	Suite 101	Lille
3	Elizabeth	Devon	UK Collectables, Ltd.	(171) 555-2282	Berkeley Gardens 12 Brewery	null	Liverpool
4	Michael	Rance	Daedalus Designs Imports	+33 1 46 62 7555	184, chausse de Tournai	null	NYC
5	Daniel	Murphy	Mini Wheels Co.	+33 1 46 62 7554	5557 North Pendale Street	null	Paris
6	Elizabeth	Benitez	FunGiftIdeas.com	5085552555	1785 First Street	null	Liverpool
7	Julie	Da Cunha	Reims Collectables	6505551386	59 rue de l'Abbaye	null	San Francisc
8	Daniel	Frick	Vitachrome Inc.	+33 1 46 62 7555	2678 Kingston Rd.	null	Paris
9	Paul	Rance	Daedalus Designs Imports	2155551555	184, chausse de Tournai	null	San Francisc
10	Elizabeth	Yu	Diecast Classics Inc.	5085552555	7586 Pompton St.	null	Liverpool
11	Martine	Rance	Daedalus Designs Imports	2155551555	184, chausse de Tournai	Suite 101	Paris
12	Michael	Yu	Diecast Classics Inc.	6505555787	7586 Pompton St.	null	Allentown
13	Michael	Murphy	Mini Wheels Co.	+33 1 46 62 7554	5557 North Pendale Street	null	NYC
14	Paul	Yu	Diecast Classics Inc.	6505551386	7586 Pompton St.	null	Reims
15	Daniel	Rance	Daedalus Designs Imports	+33 1 46 62 7554	184, chausse de Tournai	null	Lille

- Once the customer's full data is now loaded to the snowflake customer table. Let’s capture the data of the few records for which we will change the data in the next file and perform the SCD1 logic. CONTACTFIRSTNAME and CONTACTLASTNAME is the primary key for this table

- For CONTACTFIRSTNAME = 'Elizabeth' AND CONTACTLASTNAME ='Yu' we will update the PHONE, so let’s capture the current value

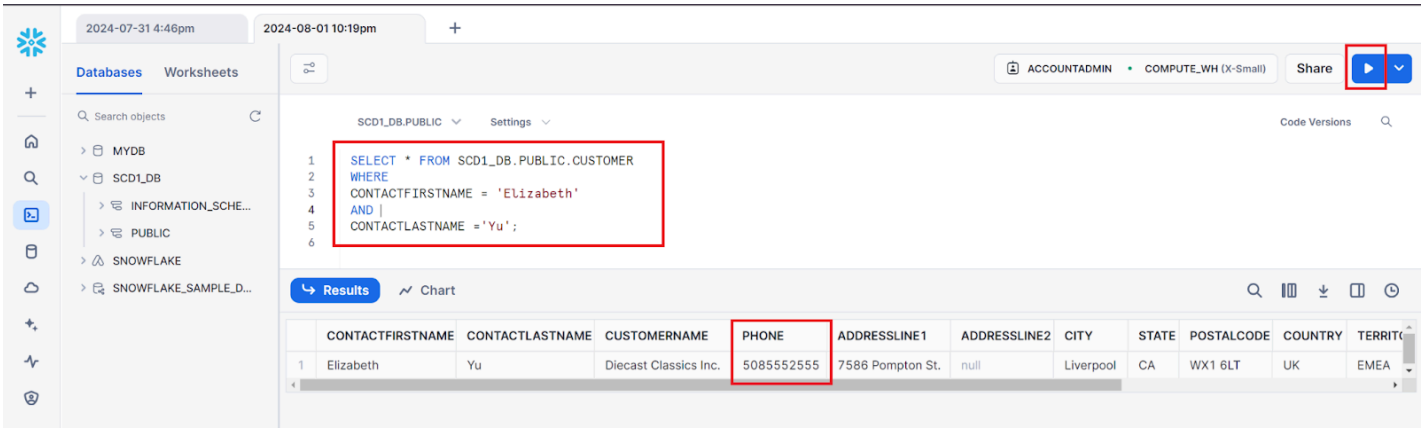
```
SELECT * FROM SCD1_DB.PUBLIC.CUSTOMER

WHERE

CONTACTFIRSTNAME = 'Elizabeth'

AND

CONTACTLASTNAME ='Yu';
```



- For CONTACTFIRSTNAME = 'Kyung' AND CONTACTLASTNAME ='Benitez' we will update the CITY, so let's capture the current value

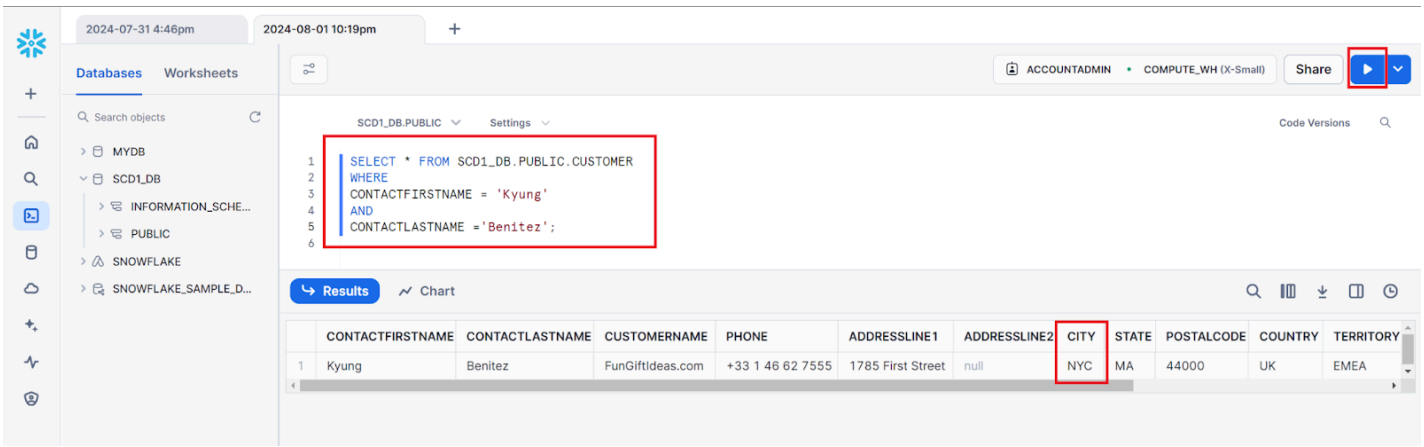
```
SELECT * FROM SCD1_DB.PUBLIC.CUSTOMER

WHERE

CONTACTFIRSTNAME = 'Kyung'

AND

CONTACTLASTNAME ='Benitez';
```



- We will add new record for CONTACTFIRSTNAME = 'John' AND CONTACTLASTNAME ='Wick', so you should get zero record by running the below query

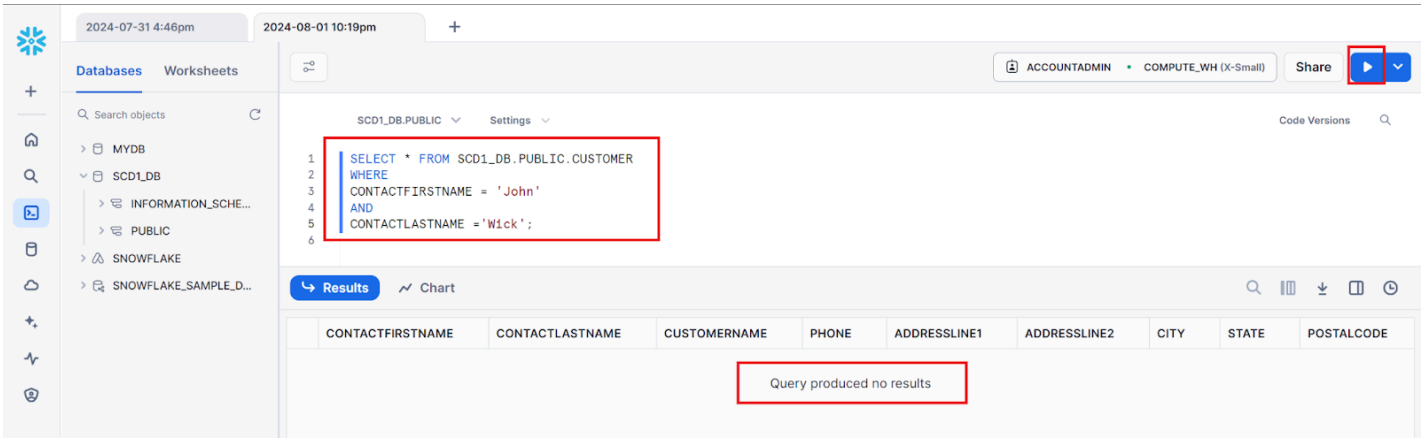
```
SELECT * FROM SCD1_DB.PUBLIC.CUSTOMER

WHERE

CONTACTFIRSTNAME = 'John'

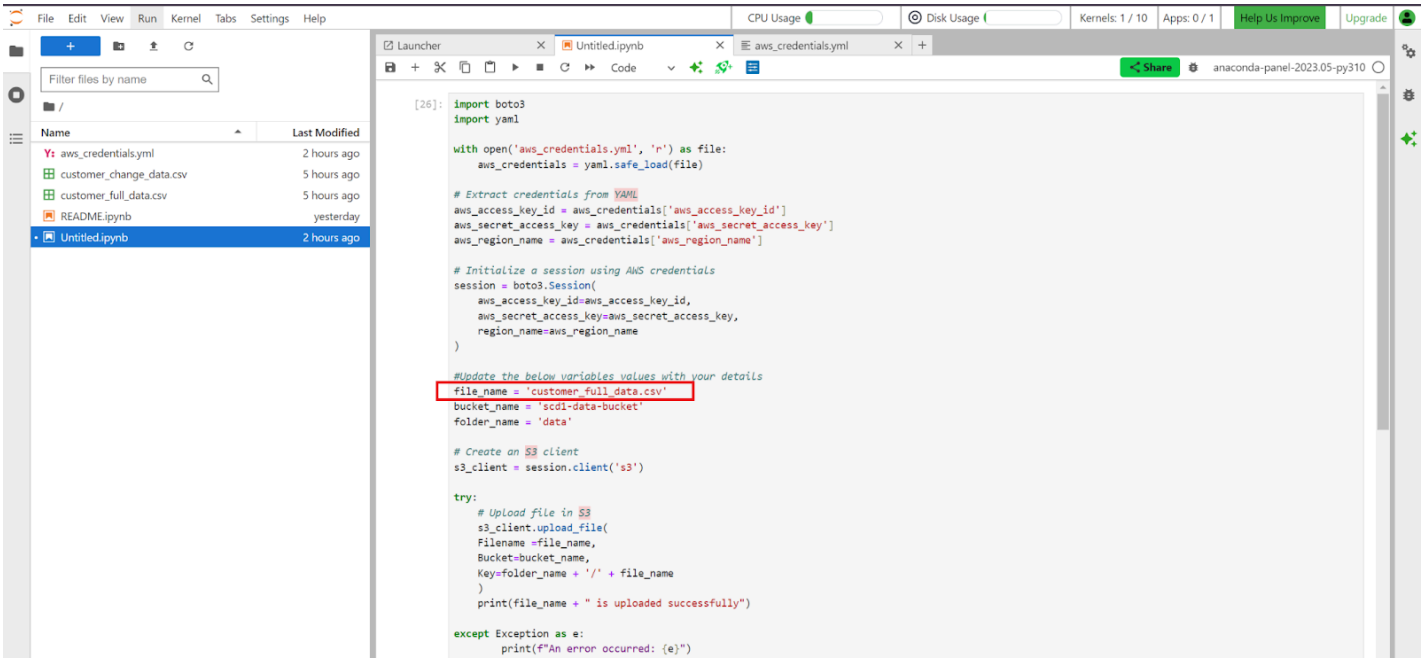
AND

CONTACTLASTNAME ='Wick';
```

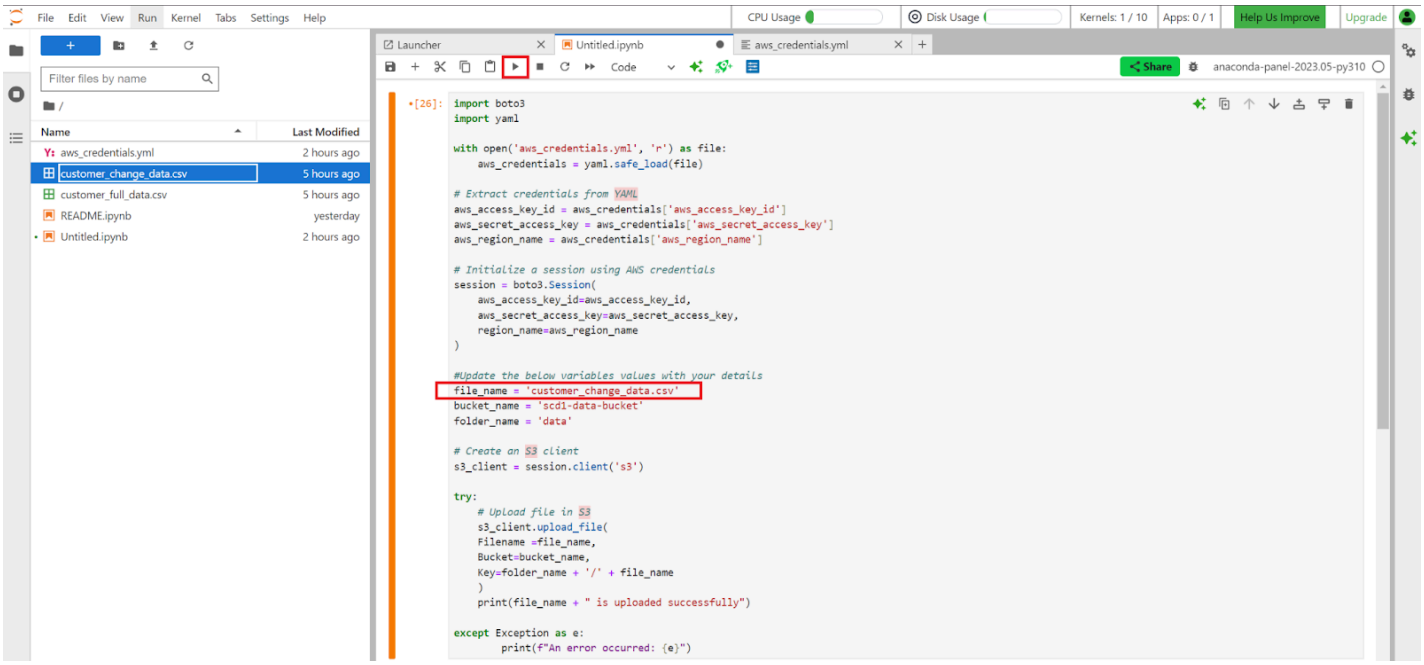


- Now let's check these changes in the customer_change_data.csv file

- Upload the customer_change_data.csv by changing the file_name variable values as highlighted below in the current python code present in the anaconda notebook

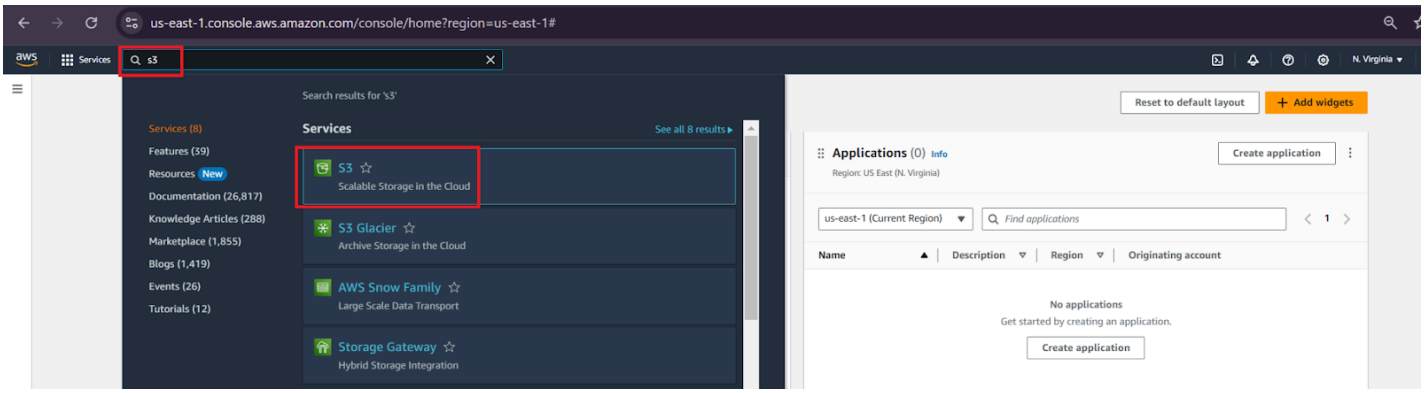


- After updating the values of the variables based on your details, click on the run button present on the top menu of your notebook to upload the customer_change_data.csv file in your respective S3 bucket folder.

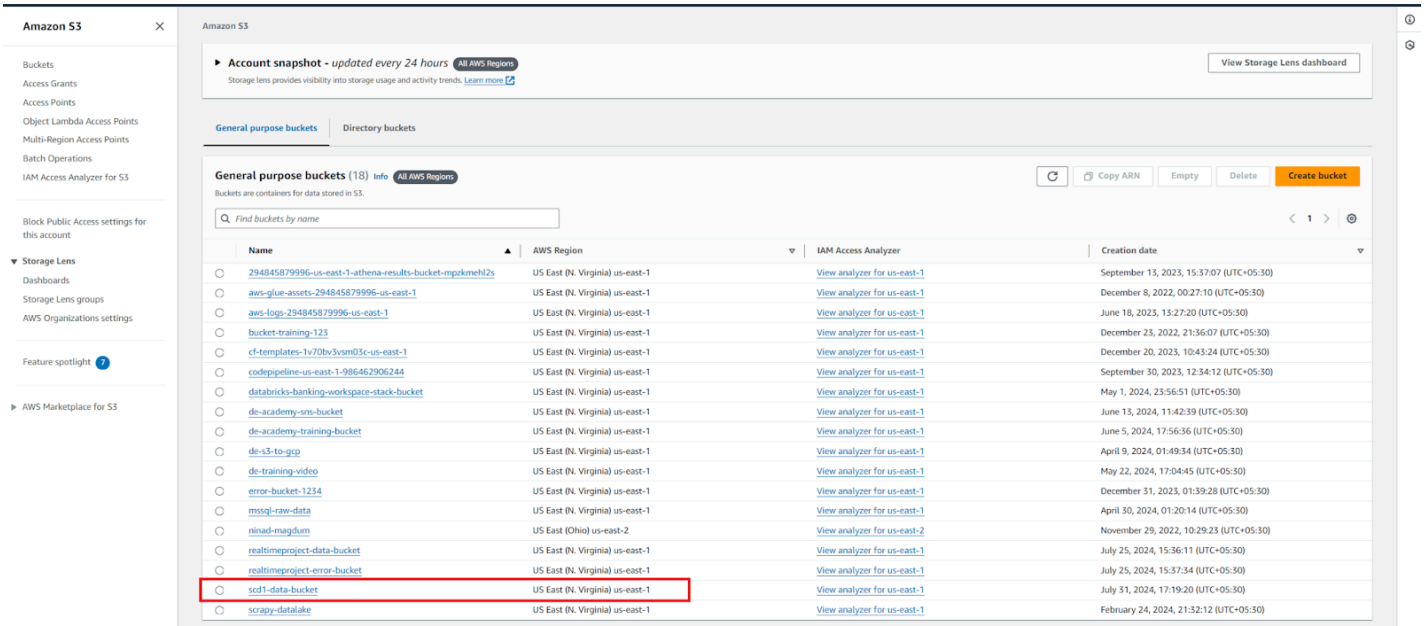


- After clicking on the run button, you should be able to see the message as customer_change_data.csv is uploaded successfully

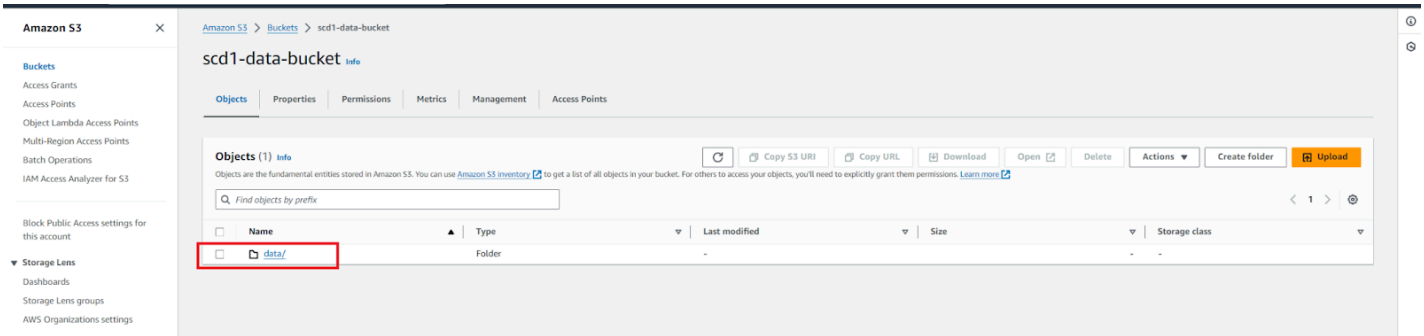
- Now the customer_change_data.csv file should be successfully uploaded to the S3 bucket folder. To validate please open the S3 bucket your AWS console by searching s3 in the search bar.



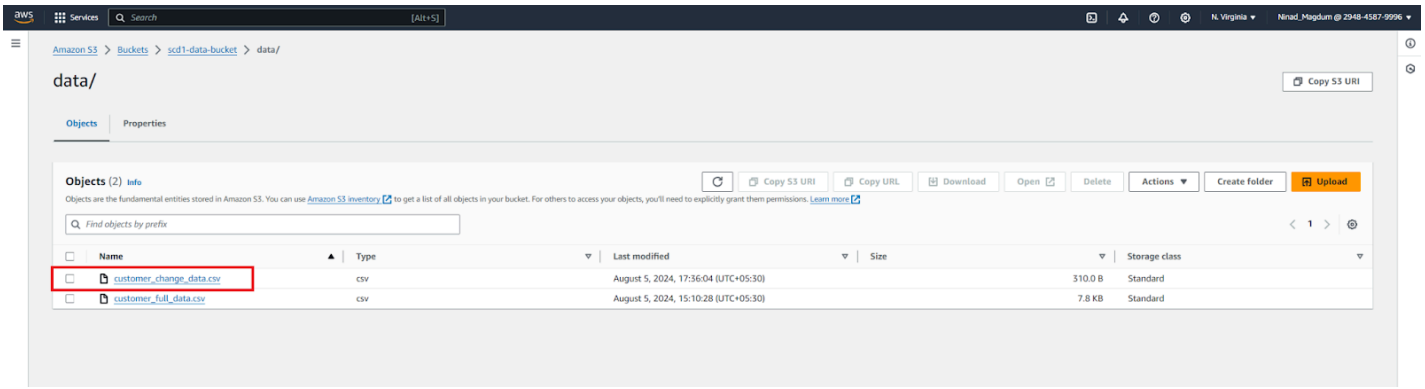
- Open you bucket once you are in the s3 bucket page



- Open the folder inside the above bucket

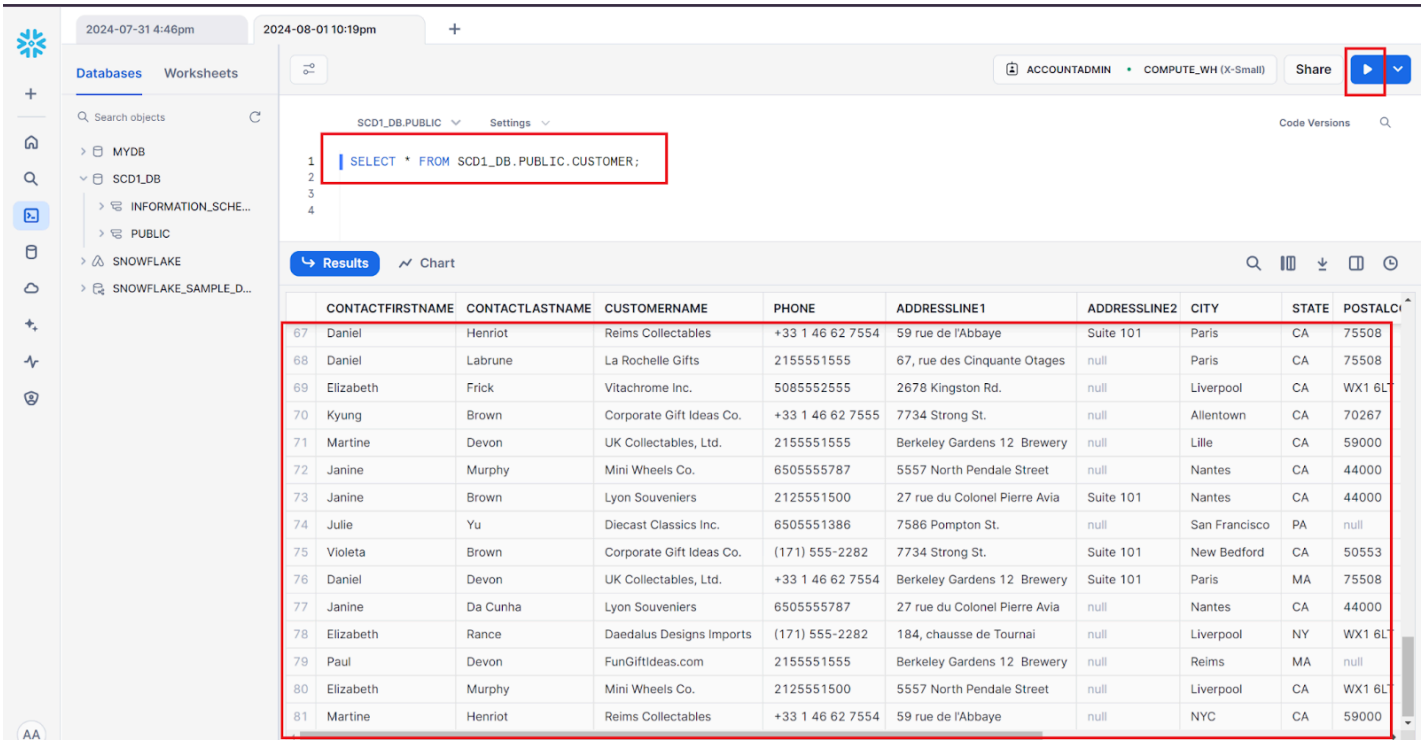


- Once you open the above folder, you should be able to see customer_change_data.csv file inside the folder.



- Now let's check the customer target table after 1 min in the snowflake to confirm if the data is now successfully loaded in the table by running the below SQL query in the snowflake worksheet. You should see 81 records.

SELECT * FROM SCD1_DB.PUBLIC.CUSTOMER;



- Once the customer's change data is now loaded to the snowflake customer table. Let's capture the data of the update records

- For CONTACTFIRSTNAME = 'Elizabeth' AND CONTACTLASTNAME ='Yu', let’s capture the current value of PHONE. The PHONE should be updated to the new value

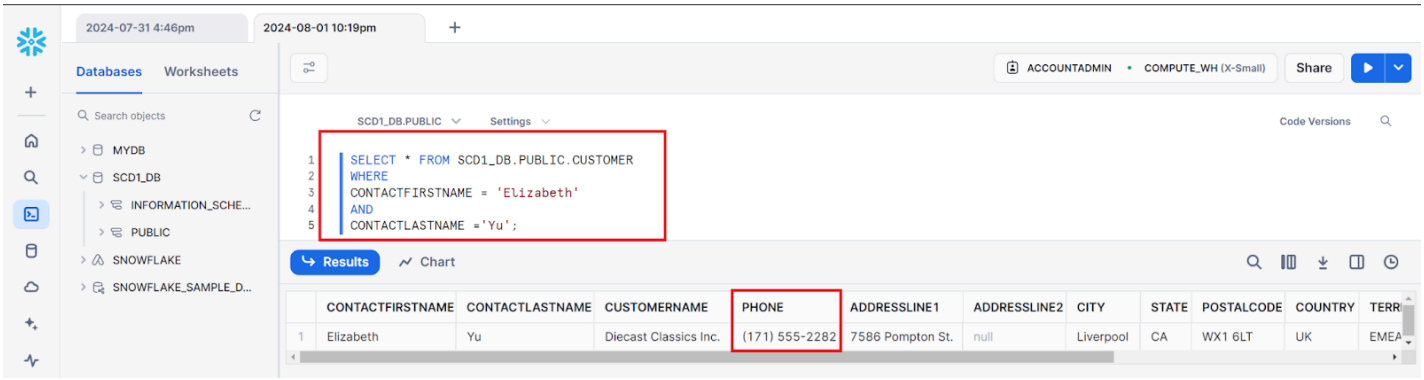
```
SELECT * FROM SCD1_DB.PUBLIC.CUSTOMER

WHERE

CONTACTFIRSTNAME = 'Elizabeth'

AND

CONTACTLASTNAME ='Yu';
```



- For CONTACTFIRSTNAME = 'Kyung' AND CONTACTLASTNAME ='Benitez' let’s capture the current value of CITY. The CITY should be updated to the new value

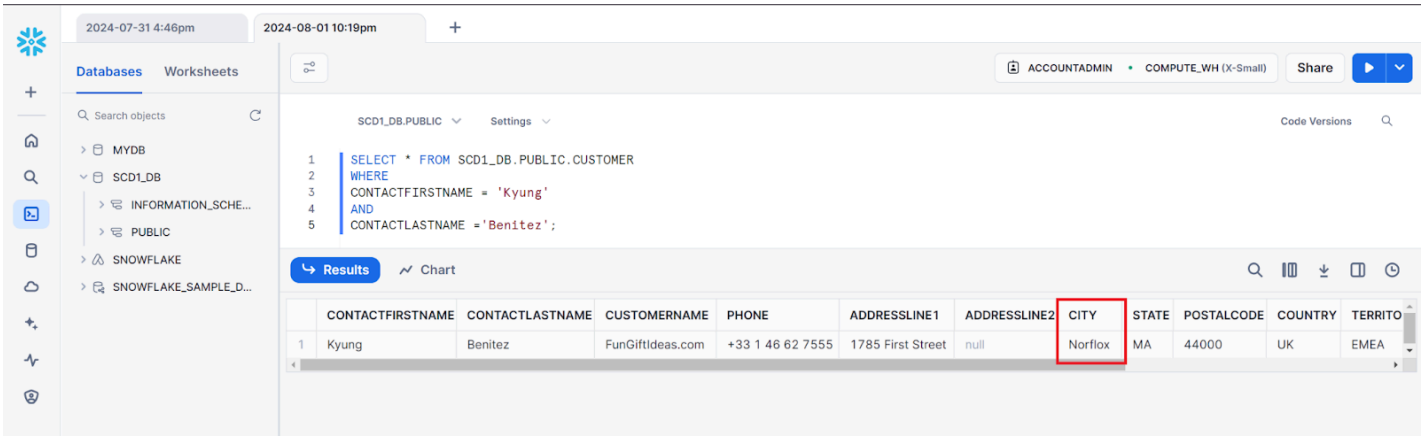
```
SELECT * FROM SCD1_DB.PUBLIC.CUSTOMER

WHERE

CONTACTFIRSTNAME = 'Kyung'

AND

CONTACTLASTNAME ='Benitez';
```



- For CONTACTFIRSTNAME = 'John' AND CONTACTLASTNAME ='Wick', You should be having a record in the table

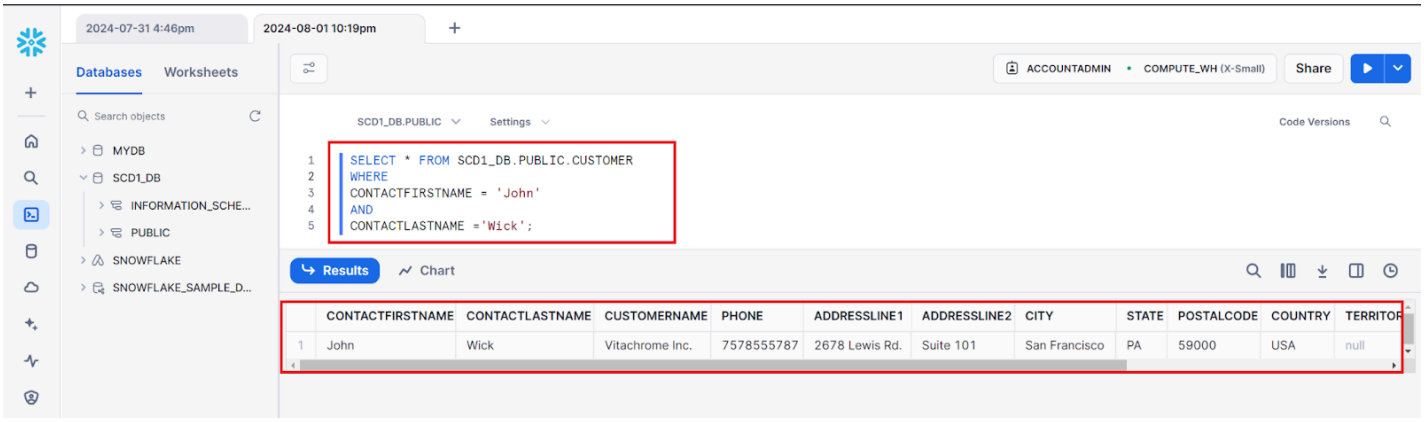
SELECT * FROM SCD1_DB.PUBLIC.CUSTOMER

WHERE

CONTACTFIRSTNAME = 'John'

AND

CONTACTLASTNAME ='Wick';



Successfully demonstrated SCD Type1

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